RAKESH KUMAR

Mechanical Engineer | CFD & Thermal-Fluid Systems Specialist

Assistant Manager – Research & Development (R&D), Metal Power Analytical Pvt. Ltd.

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PROFESSIONAL SUMMARY

Mechanical Engineer with over 7 years of experience in Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), turbomachinery, and product development. Currently working as Assistant Manager – R&D at Metal Power Analytical Pvt. Ltd., contributing to the design, development, thermal management, and performance optimization of Optical Emission Spectrometers (OES) and analytical instruments. Strong expertise in ANSYS Fluent, CFX, Structural Analysis, CAD modeling, UDF development, thermal-fluid simulations, and experimental validation. Published multiple SCI journal papers, holder of an Indian patent, and experienced in translating advanced research into industrial product development. Postdoctoral from IIT Bombay and PhD from IIT (ISM) Dhanbad.

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Manager – R&D

Metal Power Analytical Pvt. Ltd. | Mumbai, India | July 2026 – Present

- Research and development of Optical Emission Spectrometer (OES) systems.
- Thermal management and cooling design of high-temperature analytical instruments.
- CFD and thermal analysis for heat dissipation and temperature control.
- Mechanical design improvements for reliability and manufacturability.

Post-Doctoral Fellow

Indian Institute of Technology Bombay | Mumbai, India | Nov 2024 – June 2026

- Performed high-fidelity aerodynamic CFD analysis of advanced rotorcraft configurations, including cyclorotors and tandem rotors, using ANSYS Fluent, focusing on thrust generation, wake interaction, and unsteady flow behaviour.
- Developed and implemented User-Defined Functions (UDFs) for time-varying blade kinematics and complex flow physics to improve simulation accuracy.
- Applied dynamic, overset, and sliding mesh techniques to simulate rotating machinery motion and rotor-wake interactions.
- Executed HPC-based parallel CFD simulations and post-processed aerodynamic loads, pressure distribution, and flow structures for performance optimisation.

Senior Researcher

Indian Institute of Technology (Indian School of Mines) Dhanbad | Dhanbad, India | May 2024 – Nov 2024

- Wind Turbine R&D: Led the design and numerical simulation of a utility-scale wind turbine model using ANSYS CFX for performance analysis and ANSYS Static Structural for stress validation.
- Focused on developing a predictive condition monitoring system for wind turbines using simulation data.

Senior Research Fellow

Indian Institute of Technology (Indian School of Mines) Dhanbad | Dhanbad, India | Oct 2019 – Nov 2022

- Hydraulic Turbine R&D: Conducted a high-fidelity erosion and wear prediction model for radial hydraulic turbines using CFD–DEM coupling (ANSYS Fluent + EDEM).
- Analysed the influence of particle size, shape, concentration, and flow rate on turbine component erosion.

TECHNICAL SKILLS

CFD Software: ANSYS CFX & Fluent, OpenFOAM, Dynamic and Overset Mesh, Fluent Meshing (Watertight Geometry Workflow), User-Defined Functions (UDF) — turbomachinery, aerodynamics, internal flows, multiphase flow, thermal-fluid analysis, wear and erosion modelling.

Thermal & Heat Transfer: Heat transfer, thermal management, and HVAC airflow analysis.

CAD Software: SolidWorks, SpaceClaim, Siemens NX, BladeGen, AutoCAD.

FEA & FSI: ANSYS Workbench (Static & Transient Structural: Linear & Non-Linear Analysis), CFD–FEA coupling.

CFD-DEM Modelling: ANSYS Fluent–EDEM coupling; EDEM bulk/granular material simulation (wear & erosion).

HPC & Parallel Computing: Linux-based HPC clusters, MPI parallelisation, multi-node Fluent/CFX simulations, batch job submission, solver performance optimisation.

Coding: Python, MATLAB Simulink, UDF scripting, CEL (CFX Expression Language).

Post-Processing: CFD-Post, Tecplot, ParaView, Excel, OriginPro; design optimisation and parametric CFD studies.

CAE Pre-processing: ANSA (geometry cleanup, surface and volume mesh generation).

EDUCATION QUALIFICATIONS

- **Doctor of Philosophy (Ph.D.)**, Mechanical Engineering
  Indian Institute of Technology (Indian School of Mines) Dhanbad | Dhanbad, India | July 2018 - April 2024
  Thesis title: *Performance analysis of Darrieus hydrokinetic turbine in different flow conditions*.
  Date Awarded: 04 April 2024
- **Master of Technology (M.Tech.)**, Mechanical Engineering (Thermal Engineering)
  Indian Institute of Technology (Indian School of Mines) Dhanbad, | Dhanbad, India | August 2016 - May 2018
  Thesis title: *Effect of number of blades on performance of helical Darrieus hydrokinetic turbine.*
- **Bachelor of Technology (B.Tech.)**, Mechanical Engineering
  Maharana Pratap Engineering College Kanpur | AKTU Lucknow, India | July 2010 - June 2014
  Thesis title: *Design analysis and processes involved in manufacturing of a Bicycle*.

ACHIEVEMENTS

- **Award:** Inder Mohan Thapar Research (IMTR) award for the Year 2023 & 2024—Awarded by Indian Institute of Technology (Indian School of Mines), Dhanbad, India for excellence in research publications during the calendar years 2022 and 2023.
- **Patent:** Kumar, R. and Sarkar, S., "An In-Pipe Hydrokinetic Turbine (IPHKT) Structure" (Indian patent number 202331002558), Publication Date: 20-01-2023. First Examination Report: 10-02-2026.

AREAS OF EXPERTISE

Turbomachinery, thermal-fluid performance of rotating systems, aerodynamics, CFD, FSI, fluid dynamics, flow and thermal analysis, multiphase flow, FEA, CFD-DEM coupling, erosion modeling, thermal management systems, heat transfer, and HVAC systems.

PUBLICATIONS IN PEER-REVIEWED SCI JOURNALS/BOOK CHAPTERS/CONFERENCES

- Kumar, R., & Sarkar, S. (2022). Effect of design parameters on the performance of helical Darrieus hydrokinetic turbines. Renewable and Sustainable Energy Reviews, 162, 112431. https://doi.org/10.1016/j.rser.2022.112431 (Impact Factor: 16.3).
- Sarkar, S., Nag, A.K. and Kumar, R., (2023). "An overview of small-scale hydropower and its recent development". Renewable Energy Production and Distribution Volume 2, pp.249-314. https://doi.org/10.1016/B978-0-443-18439-0.00009-4.
- Kumar, R., and Sarkar, S. (2023). Erosion analysis of radial flow hydraulic turbine components through FLUENT-EDEM coupling. Powder Technology, 428, 118800. https://doi.org/10.1016/j.powtec.2023.118800 (Impact Factor: 4.6).
- Kumar, R., & Sarkar, S. (2023). Performance analysis of spherical-shaped Darrieus hydrokinetic turbine for an in-pipe hydropower system. Energy Conversion and Management, 294, 117600. https://doi.org/10.1016/j.enconman.2023.117600 (Impact Factor: 10.9).
- Kumar, R., Yadav, I., & Sarkar, S. (2024). Performance and Stress Analysis of Helical Darrieus Hydrokinetic Turbine. In International Conference on Industrial Problems on Machines and Mechanism (pp. 567-576). Singapore: Springer Nature Singapore. https://doi.org/10.1007/978-981-99-4270-1_56.
- Kumar, R., Nag, A. K., & Sarkar, S. (2024). Performance analysis of spherically curbed hydrokinetic turbine arranged in In-line array in a closed conduit. Renewable Energy, 232, 121110. https://doi.org/10.1016/j.renene.2024.121110 (Impact Factor: 9.1).
- Kumar, R., & Sarkar, S. (2025). Performance Evaluation of Modified Hydrokinetic Turbines in Pipe Flow with Velocity Correction Technique. Renewable Energy, 242, 122485. https://doi.org/10.1016/j.renene.2025.122485 (Impact Factor: 9.1).
- Kumar, R., Yadav, I., & Sarkar, S. (2026). Influence of solid particle size and concentration on centrifugal slurry pump performance. Australian Journal of Mechanical Engineering. https://doi.org/10.1080/14484846.2026.2659973. (Impact Factor: 1.3).

POSITIONS AND RESPONSIBILITY

- **Teaching Assistant**— Low Speed Aerodynamics and Rotary Wing Aerodynamics at IIT Bombay (Jan 2025 to present).

- **Teaching and Lab Assistant**— Turbomachinery, Fluid Mechanics, Fluid Machines, Engineering Mechanics, Engineering Graphics, and Theory of Machines at IIT (ISM) Dhanbad (August 2018 - April 2023).
- **Vice-President**, NGO "Karmajyoti", IIT (ISM) Dhanbad (Jan 2022 – Dec 2022); Volunteer (Jul 2018 – Dec 2022).
- **Member**, Hostel Executive Committee, Emerald Hostel, IIT (ISM) Dhanbad (2021–22).